

第九章 三相电路

知识结构

9.1 三相电路的基本概念

9.2 三相电路的分析

9.3 三相电路的功率



概念

应用1

应用2

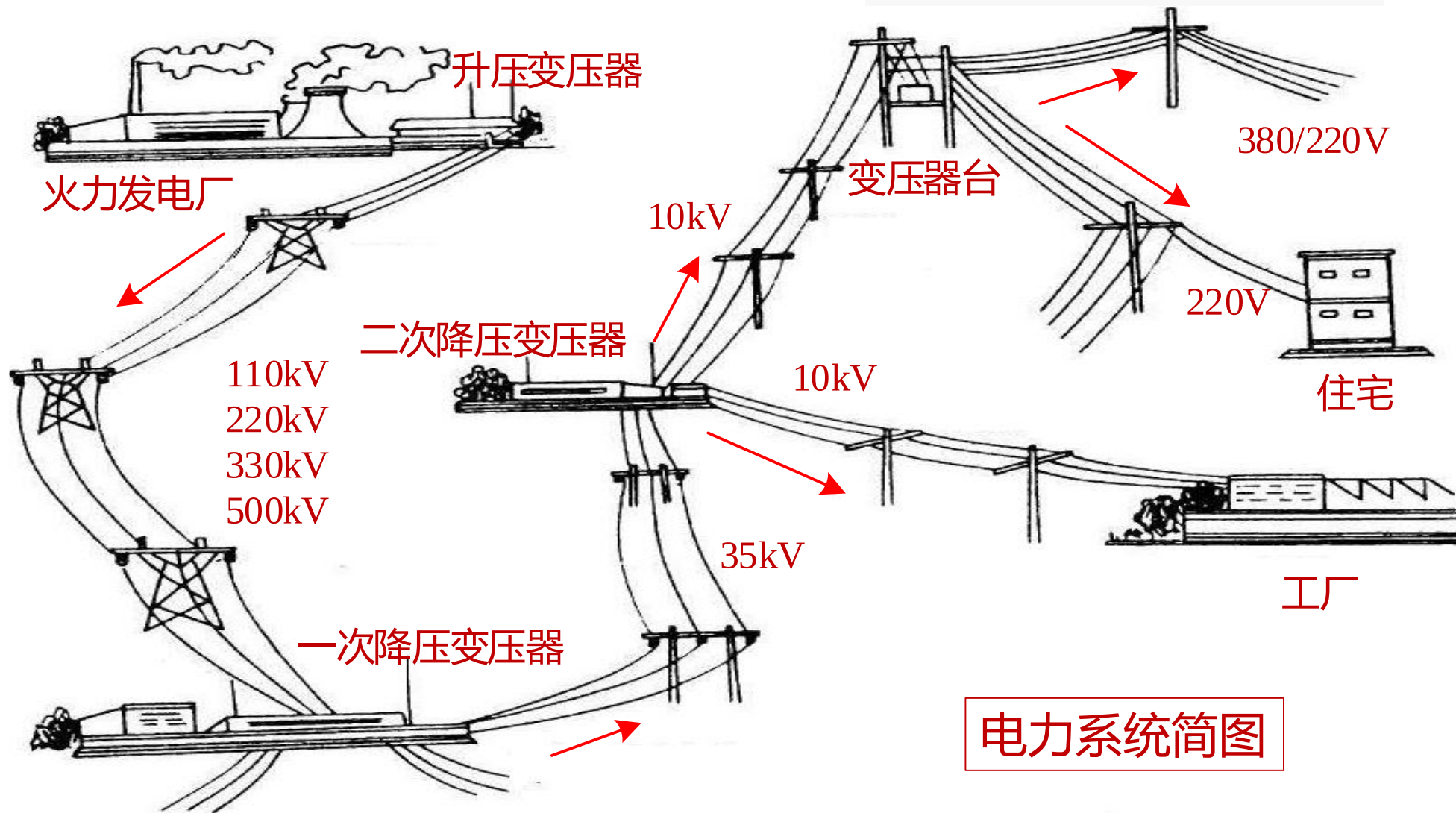
三相电路的概念:

世界首条35千伏公里级超导电缆在沪投运



新华社客户端

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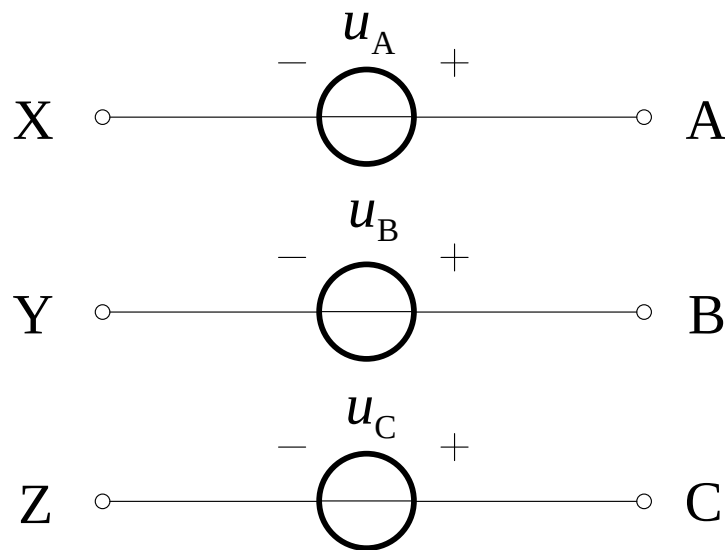
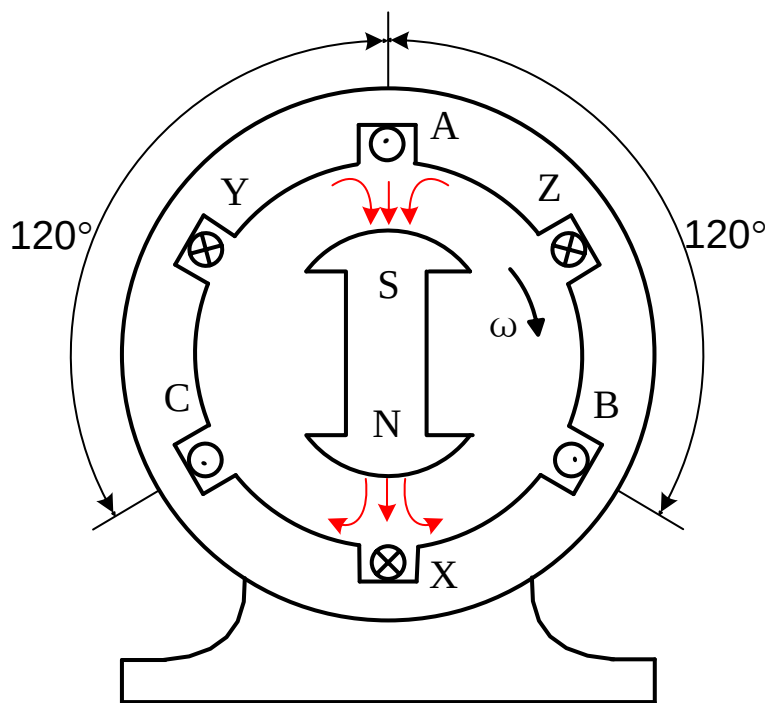
电力系统简图

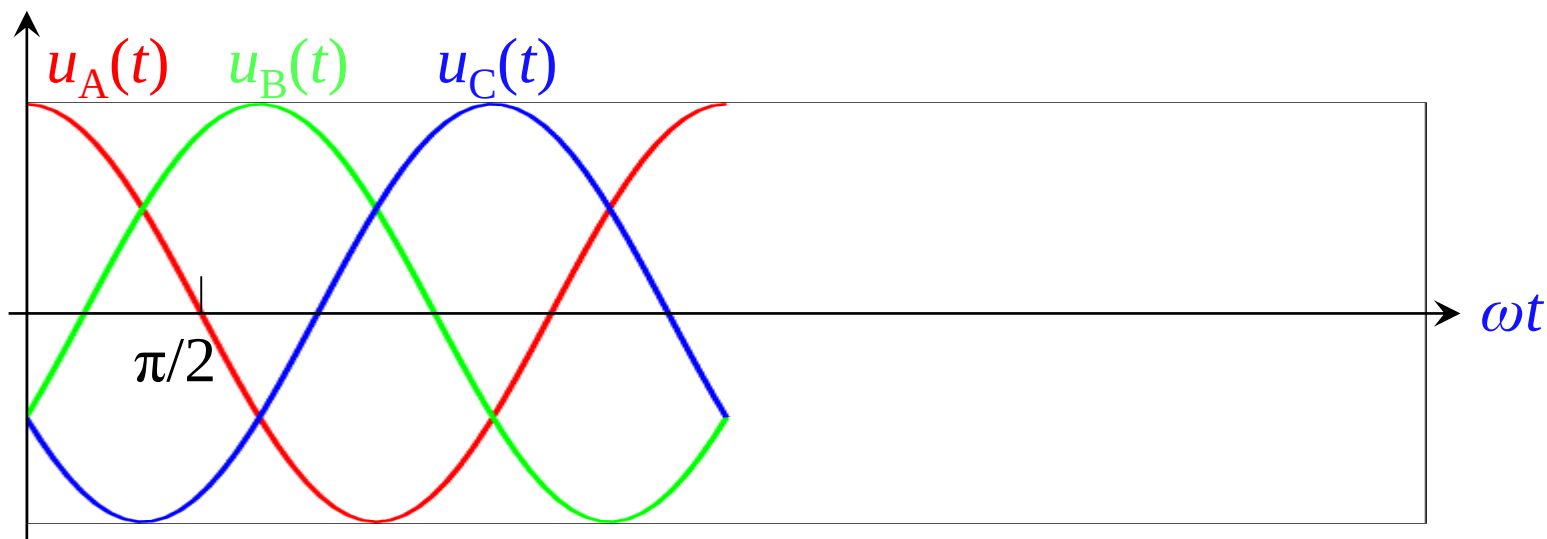
9.1 三相电路的基本概念

9.1.1 三相制

三相制：是指由三相电源供电的体系。电力系统在发电、输电和配电方面以及大功率的用电设备大多采用三相制。

三相电源：由三相发电机产生，幅值、频率相同而相位对称的电源。

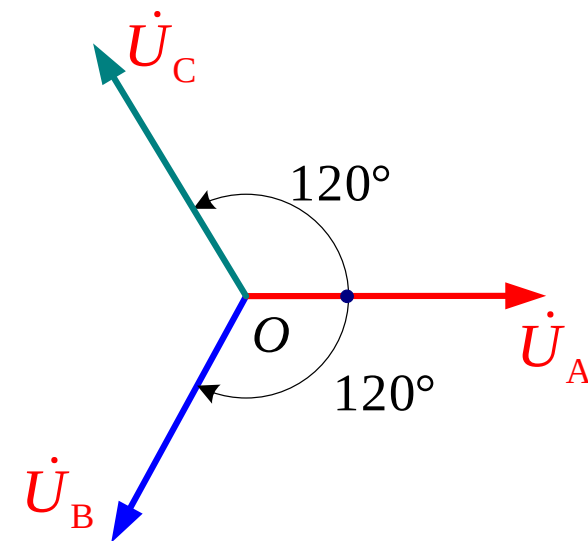




对称三相电压可表示为:

$$\begin{cases} u_A(t) = \sqrt{2}U \cos(\omega t + \varphi) \\ u_B(t) = \sqrt{2}U \cos(\omega t + \varphi - 120^\circ) \\ u_C(t) = \sqrt{2}U \cos(\omega t + \varphi + 120^\circ) \end{cases} \Rightarrow \begin{cases} \dot{U}_A = U \angle \varphi \\ \dot{U}_B = U \angle \varphi - 120^\circ \\ \dot{U}_C = U \angle \varphi + 120^\circ \end{cases}$$

一般取 $\varphi = 0$



$$\begin{cases} \dot{U}_A + \dot{U}_B + \dot{U}_C = 0 \\ u_A(t) + u_B(t) + u_C(t) = 0 \end{cases}$$

定义: 每相电压达到最大值的次序, 称为相序。正序: ABC; 负序: ACB

9.1.2 三相制的Y形接法和Δ形接法

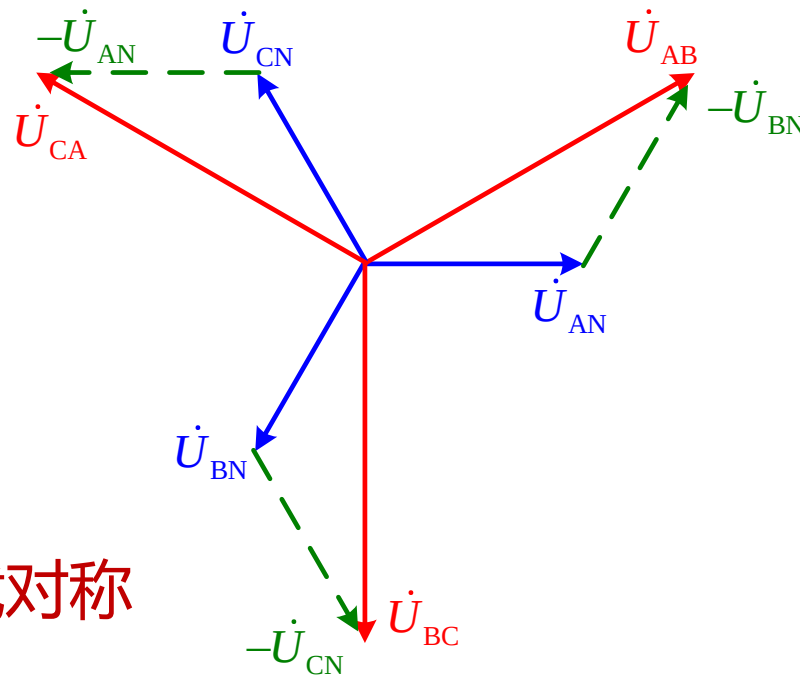
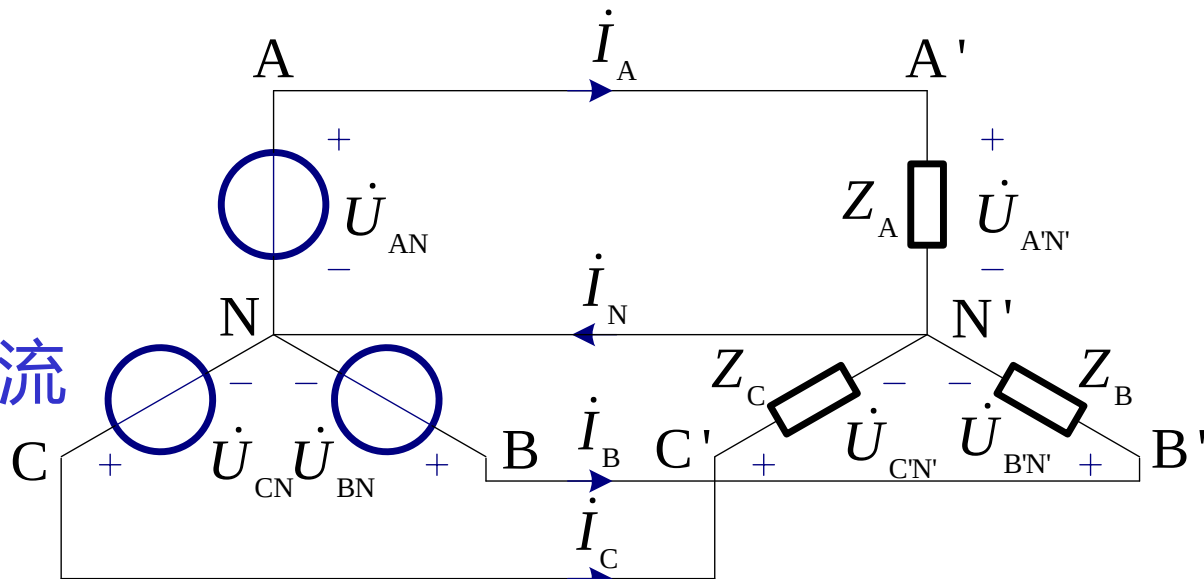
一、Y形接法

相电压/电流： 每相电源或负载的电压/电流

线电压： 端线（火线）之间的电压

线电流： 流过端线（火线）的电流

中线电流： 流过中线的电流



$$\dot{U}_{AB} = \dot{U}_A - \dot{U}_B = U_p \angle 0^\circ - U_p \angle -120^\circ$$

$$= \dot{U}_A (1 - \angle -120^\circ) = \sqrt{3} \dot{U}_A \angle 30^\circ$$

$$\dot{U}_{BC} = \sqrt{3} \dot{U}_B \angle 30^\circ$$

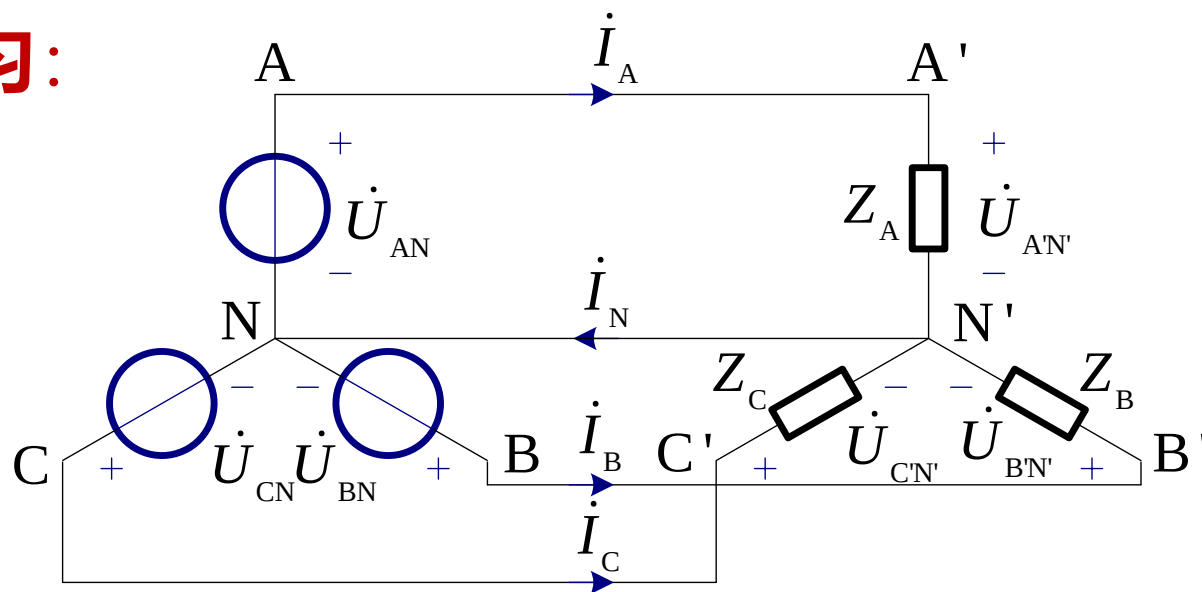
$$\dot{U}_{CA} = \sqrt{3} \dot{U}_C \angle 30^\circ$$

$$\Rightarrow U_L = \sqrt{3} U_p$$

$$I_L = I_p$$

如负载对称

练习:



相电压、相电流、线电压、线
电流、端线（火线）、中线、
中线电流

- A. 无论中线断开与否, $\dot{U}_{AB}$ 保持不变。
- B. 无论中线断开与否, $\dot{I}_A$ 保持不变。
- C. 无论三相负载对称与否, 都有 $U_L = \sqrt{3}U_p$ 。
- D. 无论三相负载对称与否, 都有 $I_L = I_p$ 。

二、 Δ 形接法

如负载对称

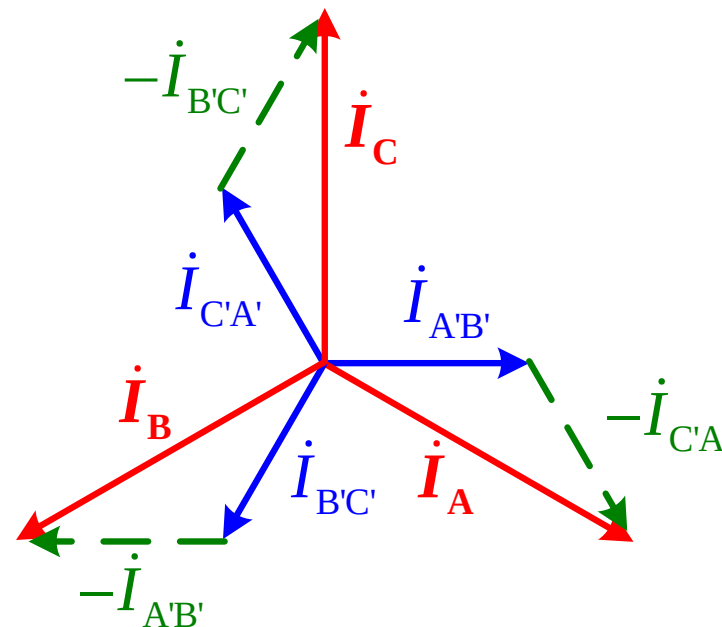
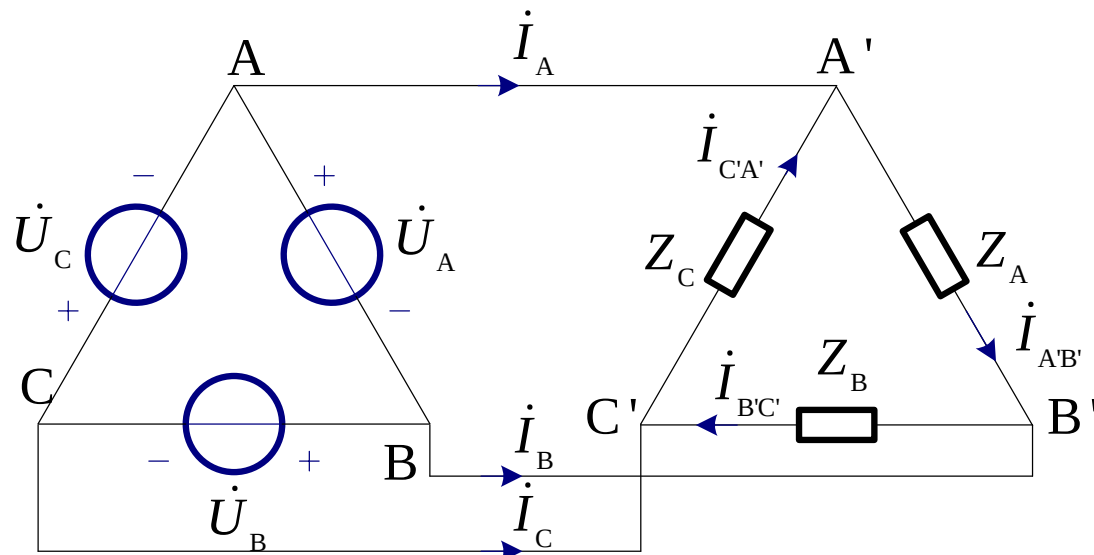
每相的极性
不能接反!

$$\begin{cases} \dot{I}_A = \dot{I}_{A'B'} - \dot{I}_{C'A'} \\ \dot{I}_B = \dot{I}_{B'C'} - \dot{I}_{A'B'} \\ \dot{I}_C = \dot{I}_{C'A'} - \dot{I}_{B'C'} \end{cases}$$

$$\begin{cases} \dot{I}_A = \sqrt{3} \angle -30^\circ \dot{I}_{A'B'} \\ \dot{I}_B = \sqrt{3} \angle -30^\circ \dot{I}_{B'C'} \\ \dot{I}_C = \sqrt{3} \angle -30^\circ \dot{I}_{C'A'} \end{cases}$$

$$I_L = \sqrt{3} I_p$$

$$U_L = U_p$$



练习：下述正确的是

- A. $\dot{U}_A + \dot{U}_B + \dot{U}_C = 0$ 成立的条件是，三相电路为正序。
- B. Δ 形连接的对称三相负载的线电流和相电流相等。
- C. 星形连接的三相电源的线电压是相电压的 $\sqrt{3}$ 倍。
- D. 星形连接的不对称三相负载的线电压是相电压的 $\sqrt{3}$ 倍。
- E. 三相电路中的对称指电路拓扑和相位都对称。

例：如图，已知 $Z = (10 + j7.54)\Omega$ ，与Y形电源相连， $\dot{U}_a = 220\angle 30^\circ \text{V}$ ，相电压为正序，试求负载相电流和线电流。

$$\dot{U}_a = 220\angle 30^\circ \text{V} \Rightarrow \dot{U}_b = 220\angle -90^\circ \text{V}, \dot{U}_c = 220\angle 150^\circ \text{V}$$

$$\Rightarrow \dot{U}_{ab} = \dot{U}_a - \dot{U}_b = 381\angle 60^\circ \text{V}$$

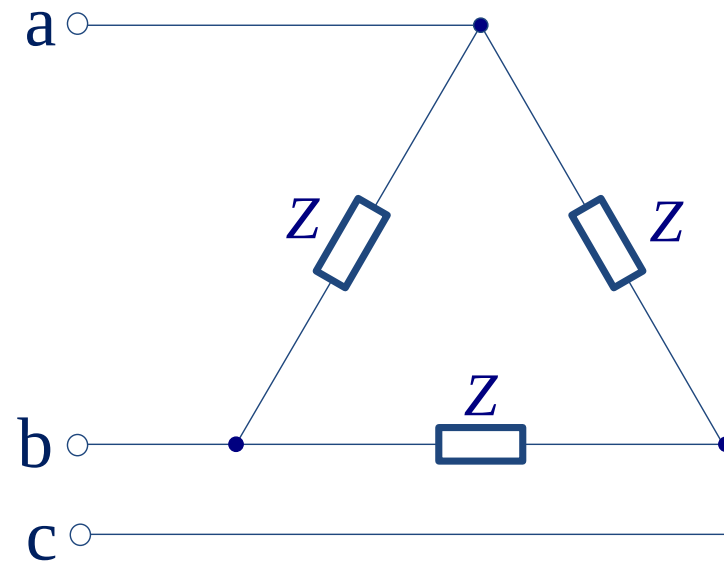
$$\dot{I}_{ab} = \dot{U}_{ab} / Z = \frac{381\angle 60^\circ}{10 + j7.54} = 30.43\angle 22.98^\circ \text{A}$$

$$\dot{I}_{bc} = 30.43\angle (22.98^\circ - 120^\circ) = 30.43\angle -97.02^\circ \text{A}$$

$$\dot{I}_{ca} = 30.43\angle (22.98^\circ + 120^\circ) = 30.43\angle 142.98^\circ \text{A}$$

$$\dot{I}_a = \dot{I}_{ab} - \dot{I}_{ca} = \sqrt{3}\dot{I}_{ab}\angle -30^\circ = 30.43\angle 22.98^\circ - 30.43\angle 142.98^\circ = 52.70\angle -7.02^\circ \text{A}$$

$$\dot{I}_b = 52.70\angle -127.02^\circ \text{A}, \dot{I}_c = 52.70\angle 112.98^\circ \text{A}$$



9.2 三相电路的分析

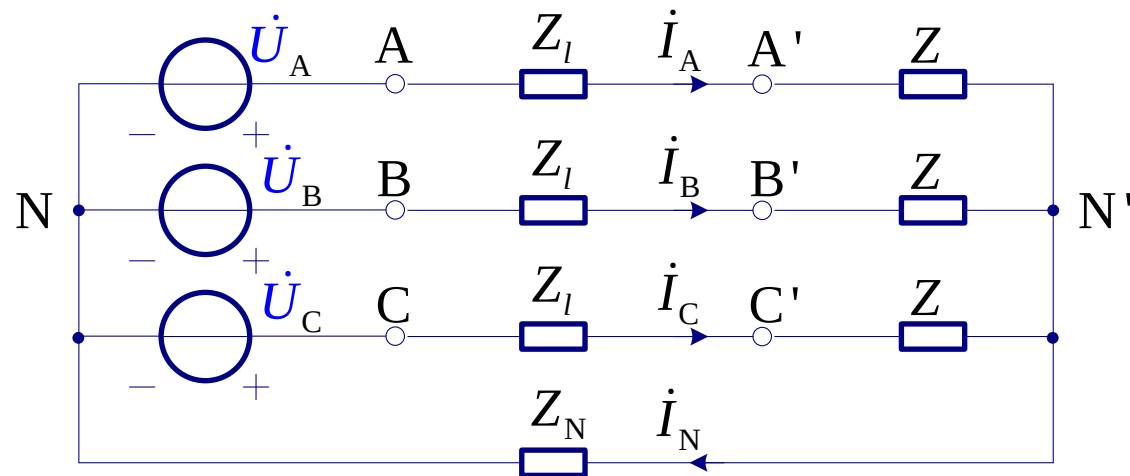
9.2.1 对称三相电路的分析

由节点法, 得

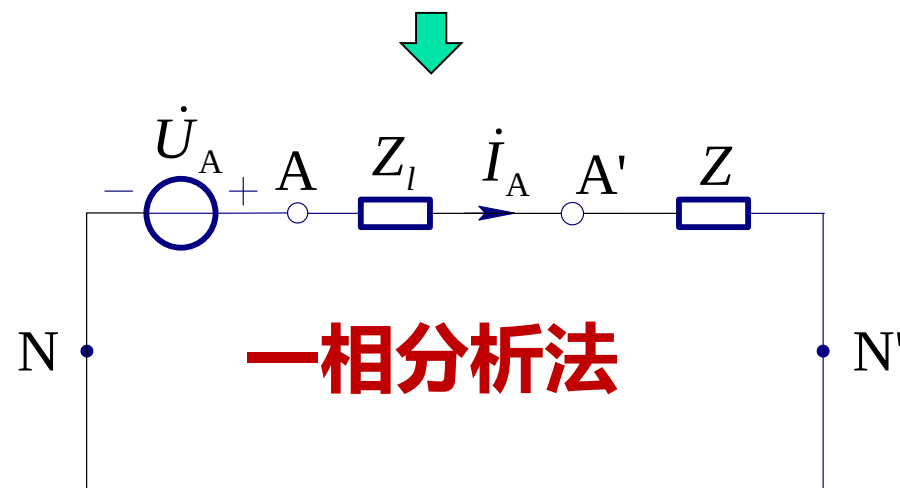
$$\left(\frac{3}{Z_l + Z} + \frac{1}{Z_N}\right)\dot{U}_{N'N} = \frac{\dot{U}_A}{Z_l + Z} + \frac{\dot{U}_B}{Z_l + Z} + \frac{\dot{U}_C}{Z_l + Z}$$

$$\Rightarrow \dot{U}_{N'N} = \frac{(\dot{U}_A + \dot{U}_B + \dot{U}_C) / (Z_l + Z)}{3 / (Z_l + Z) + 1 / Z_N} = 0$$

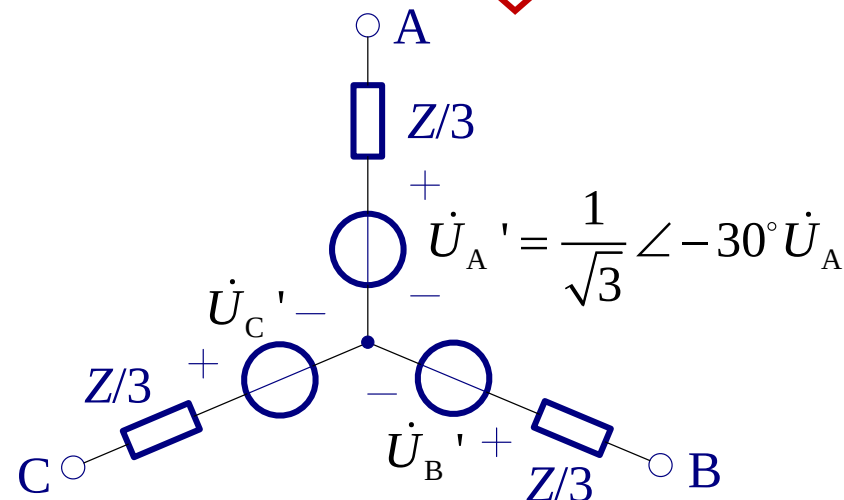
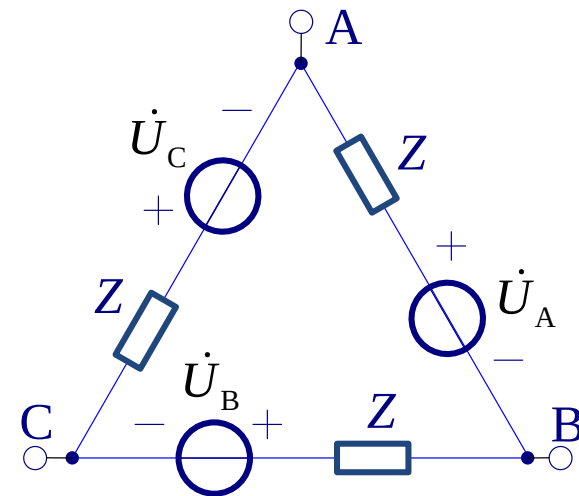
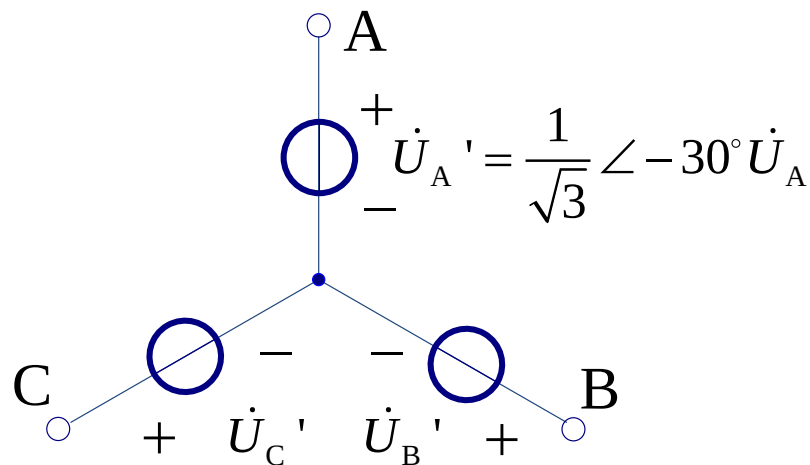
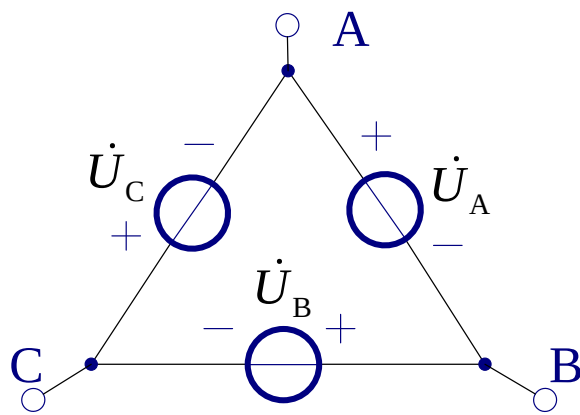
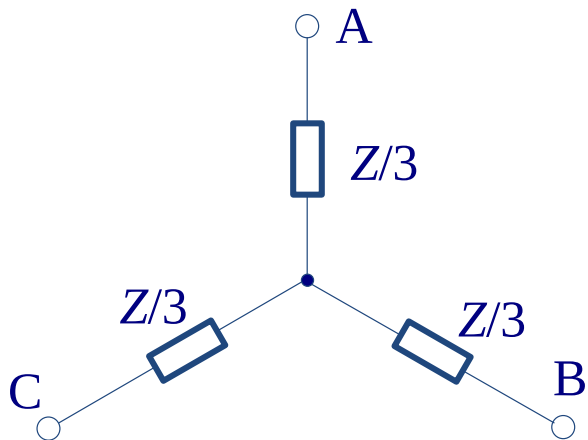
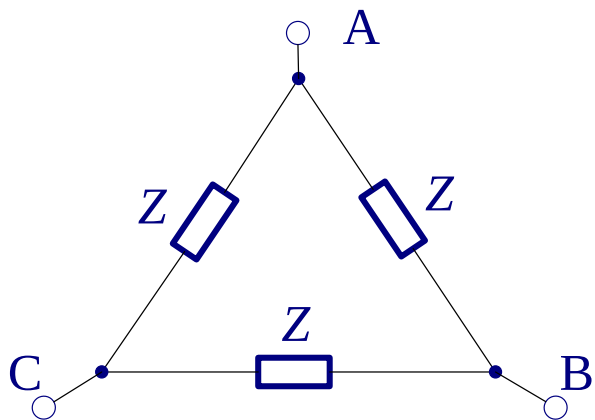
$$\left. \begin{aligned} \dot{U}_{A'N'} &= \dot{I}_A Z \\ \dot{U}_{B'N'} &= a^2 \dot{U}_{A'N'} \\ \dot{U}_{C'N'} &= a \dot{U}_{A'N'} \end{aligned} \right\} \left. \begin{aligned} \dot{I}_A &= \dot{U}_A / (Z_l + Z) \\ \dot{I}_B &= \dot{I}_A \angle -120^\circ \\ &= \dot{I}_A (\angle 120^\circ)^2 = a^2 \dot{I}_A \\ \dot{I}_C &= a \dot{I}_A \end{aligned} \right\}$$

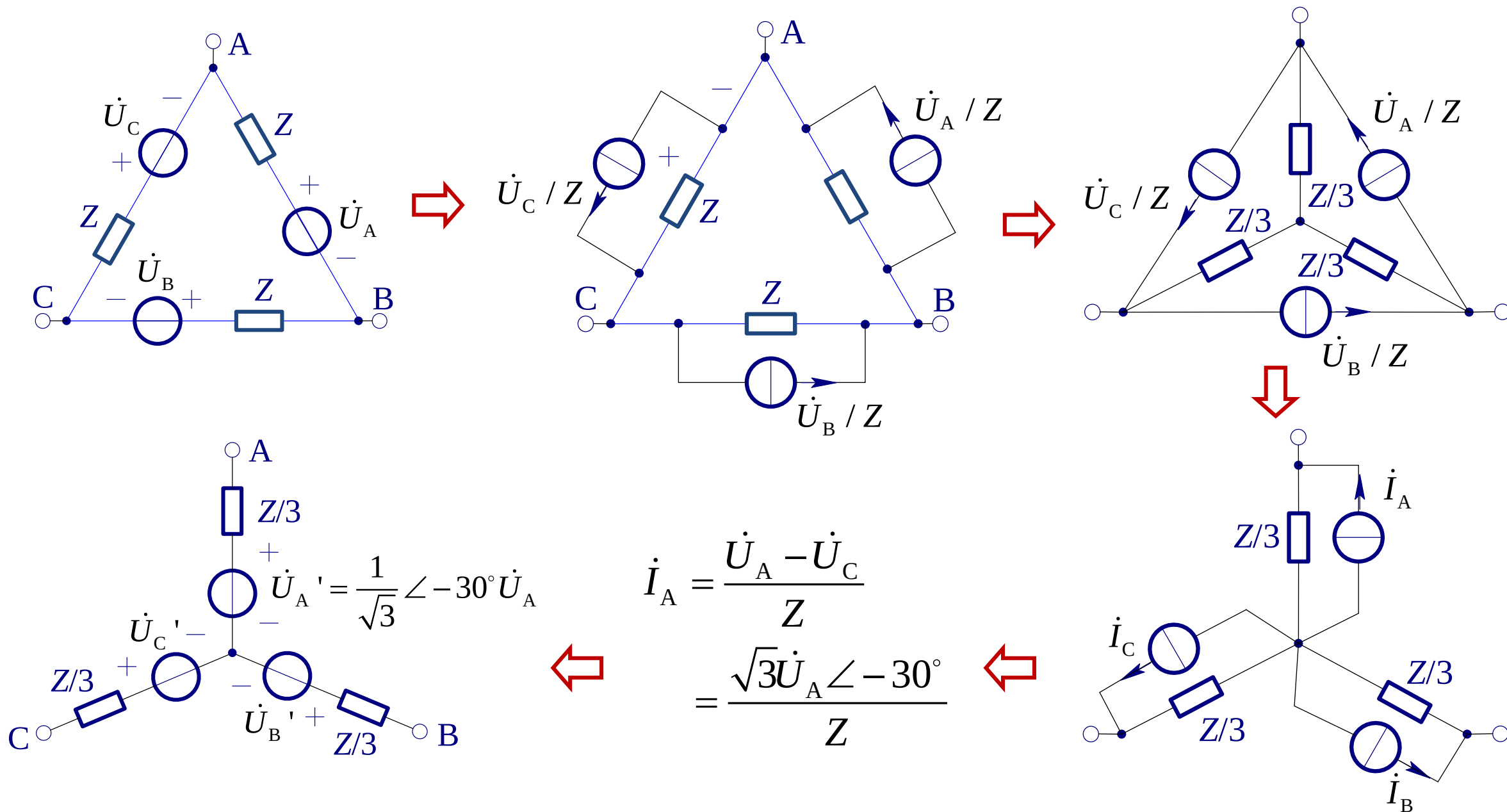


可取消中线
三相四线制 → 三相三线制



其它形式的三相电路:





例：如图，已知 $Z=(5+j4)\Omega$ ， $Z_l=(0.5+j0.4)\Omega$ ，端点A、B、C接于对称三相电源，电源电压为380V。试求 $\dot{I}_{A1}$ 、 $\dot{I}_{A2}$ 、 $\dot{I}_A$ 。

$$\dot{U}_{AN} = 220\angle 0^\circ \text{ V}$$

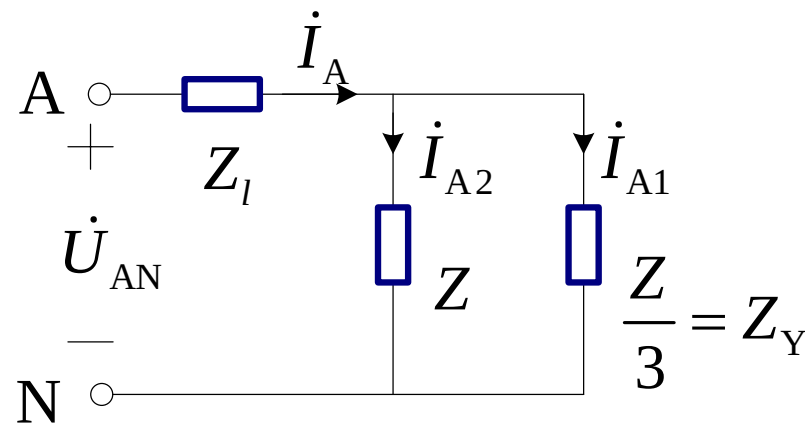
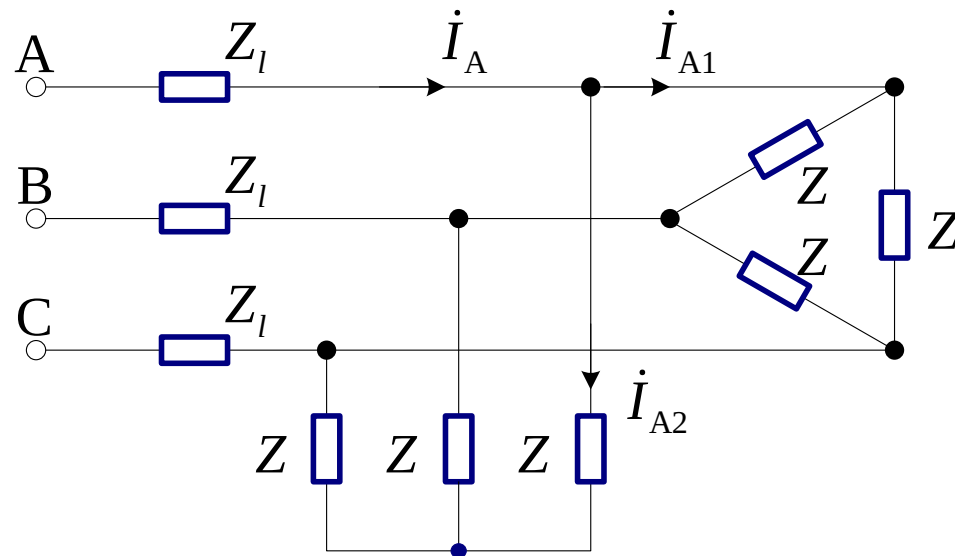
$$Z_Y = \frac{Z}{3} = \left(\frac{5}{3} + j\frac{4}{3}\right)\Omega$$

$$Z_A = Z_l + \frac{1/3}{1+1/3} Z = (1.75 + j1.4)\Omega$$

$$\dot{I}_A = \frac{\dot{U}_{AN}}{Z_A} = \frac{220\angle 0^\circ}{1.75 + j1.4} = 98.17\angle -38.66^\circ \text{ A}$$

$$\dot{I}_{A1} = 3\dot{I}_A / 4 = 73.63\angle -38.7^\circ \text{ A}$$

$$\dot{I}_{A2} = \dot{I}_A / 4 = 24.54\angle -38.7^\circ \text{ A}$$

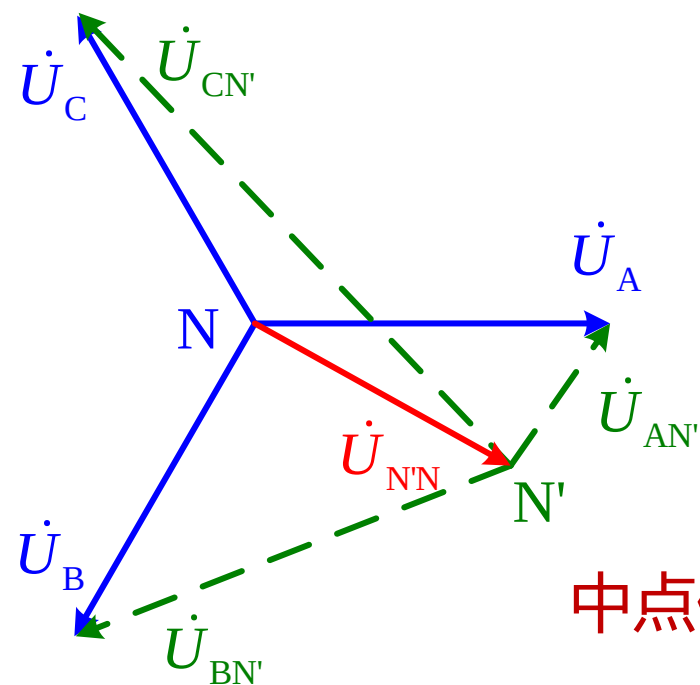
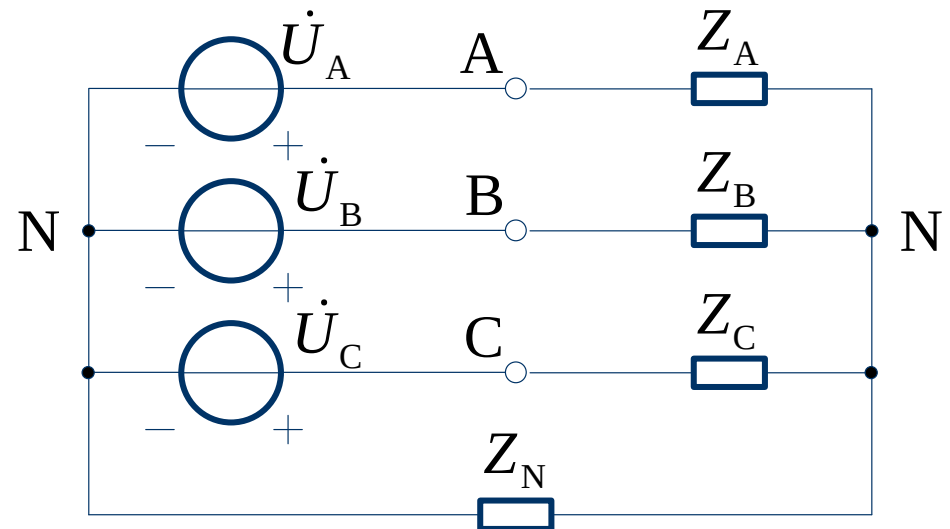


9.2.2 不对称三相电路的分析

$$\dot{U}_{N'N} = \frac{\frac{\dot{U}_A}{Z_A} + \frac{\dot{U}_B}{Z_B} + \frac{\dot{U}_C}{Z_C}}{\frac{1}{Z_A} + \frac{1}{Z_B} + \frac{1}{Z_C} + \frac{1}{Z_N}}$$

$$\begin{cases} \dot{U}_{AN'} = \dot{U}_A - \dot{U}_{N'N} \\ \dot{U}_{BN'} = \dot{U}_B - \dot{U}_{N'N} \\ \dot{U}_{CN'} = \dot{U}_C - \dot{U}_{N'N} \end{cases}$$

分析方法：一般正弦稳态电路分析



中点位移

例：已知 $Z_A = 10\angle 30^\circ \Omega$, $Z_B = 20\angle 60^\circ \Omega$, $Z_C = 15\angle -45^\circ \Omega$, 线电压有效值为440V, 负相序。求线电流、中线电流。

$$\dot{U}_A = \frac{U_l}{\sqrt{3}} \angle 0^\circ = \frac{440}{\sqrt{3}} \angle 0^\circ = 254 \angle 0^\circ \text{ V}$$

$$\dot{U}_B = 254 \angle (0^\circ + 120^\circ) = 254 \angle 120^\circ \text{ V}$$

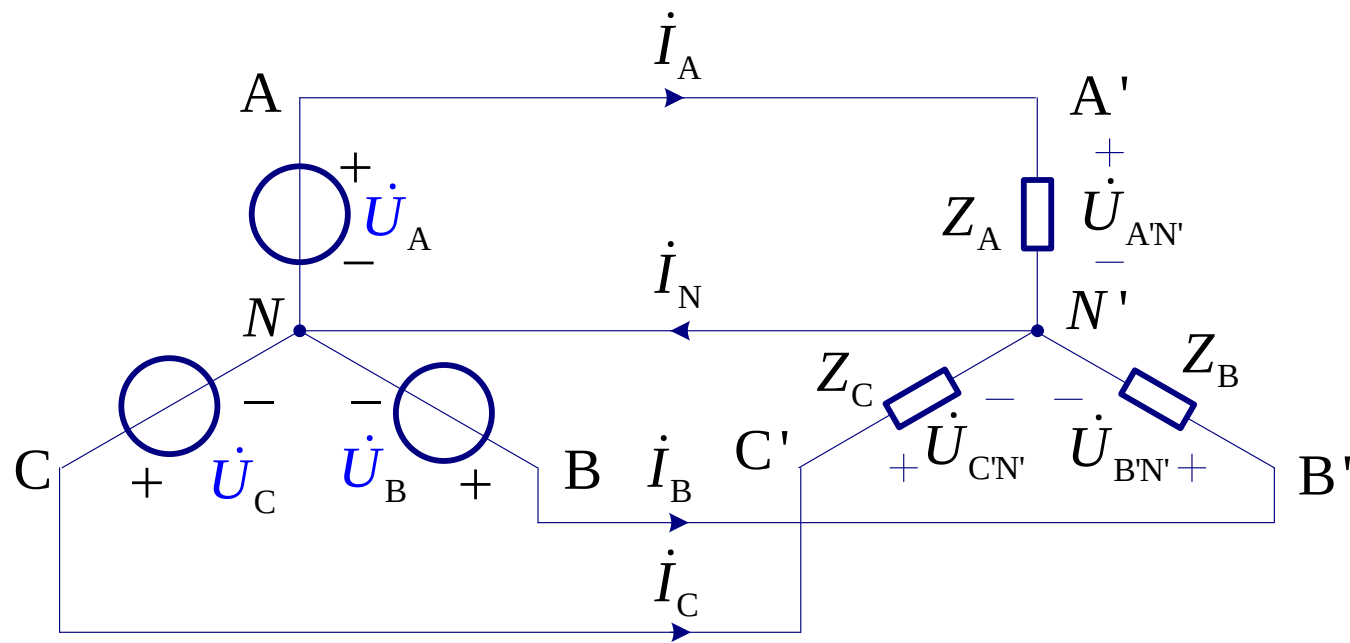
$$\dot{U}_C = 254 \angle -120^\circ \text{ V}$$

$$\dot{I}_A = \dot{U}_A / Z_A = 25.4 \angle -30^\circ \text{ A}$$

$$\dot{I}_B = \dot{U}_B / Z_B = 16.93 \angle -75^\circ \text{ A}$$

$$\dot{I}_C = \dot{U}_C / Z_C = 12.70 \angle 60^\circ \text{ A}$$

$$\dot{I}_N = \dot{I}_A + \dot{I}_B + \dot{I}_C = 37.38 \angle -28.88^\circ \text{ A}$$



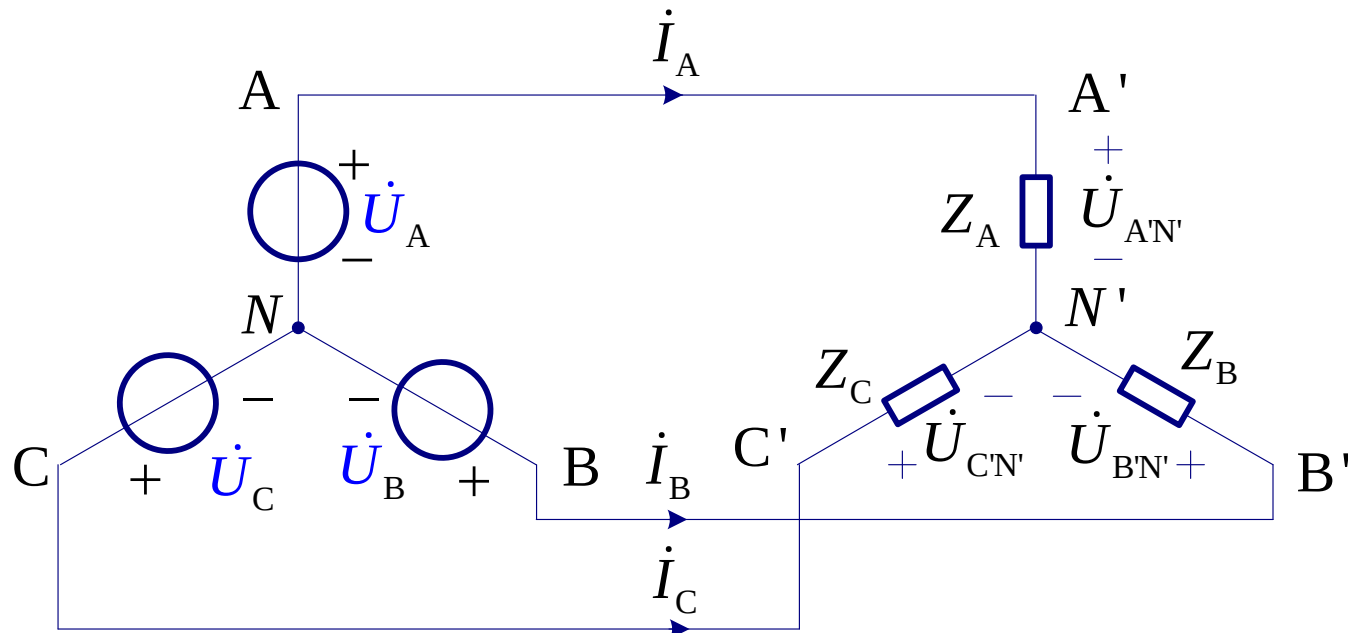
例：接上例。取消中线，求线电流及 $\dot{U}_{N'N}$ 。

$$\dot{U}_A = 254\angle 0^\circ \text{ V} \quad Z_A = 10\angle 30^\circ \Omega$$

$$\dot{U}_B = 254\angle 120^\circ \text{ V} \quad Z_B = 20\angle 60^\circ \Omega$$

$$\dot{U}_C = 254\angle -120^\circ \text{ V} \quad Z_C = 15\angle -45^\circ \Omega$$

$$\begin{aligned} (1/Z_A + 1/Z_B + 1/Z_C)\dot{U}_{N'N} \\ = \dot{U}_A/Z_A + \dot{U}_B/Z_B + \dot{U}_C/Z_C \end{aligned}$$



$$\Rightarrow \dot{U}_{N'N} = 226.1\angle -12.67^\circ \text{ V} \quad \Rightarrow \begin{cases} \dot{I}_A = (\dot{U}_A - \dot{U}_{N'N}) / Z_A = 5.98\angle 26.05^\circ \text{ A} \\ \dot{I}_B = 21.99\angle 82.21^\circ \text{ A} \\ \dot{I}_C = 25.81\angle -108.89^\circ \text{ A} \end{cases}$$

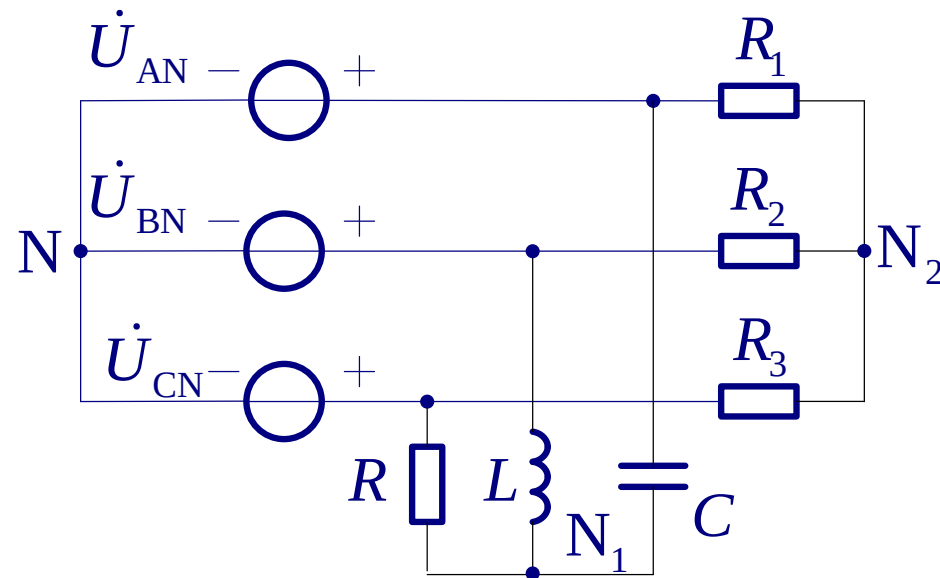
例：如图，电源为对称三相电压源，已知 $\dot{U}_{AN} = 10\angle 0^\circ \text{V}$ ， $R=10\Omega$ ， $\omega L=10\Omega$ ， $(1/\omega C)=10\Omega$ ， $R_1=R_2=R_3=10\Omega$ 。试 $\dot{U}_{N_2N_1}$ 。

$$\begin{aligned} (1/Z_A + 1/Z_B + 1/Z_C)\dot{U}_{N'N} \\ = \dot{U}_A/Z_A + \dot{U}_B/Z_B + \dot{U}_C/Z_C \end{aligned}$$

$$\begin{cases} \left(\frac{1}{10} + \frac{1}{j10} + \frac{1}{-j10}\right)\dot{U}_{N_1N} = \frac{10\angle 0^\circ}{-j10} + \frac{10\angle -120^\circ}{j10} + \frac{10\angle 120^\circ}{10} \\ \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{10}\right)\dot{U}_{N_2N} = \frac{10\angle 0^\circ}{10} + \frac{10\angle -120^\circ}{10} + \frac{10\angle 120^\circ}{10} = 0 \end{cases}$$

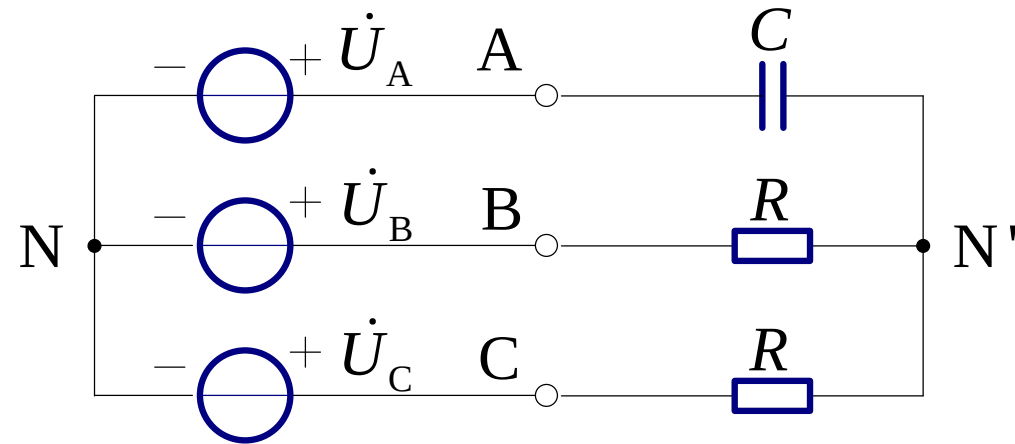
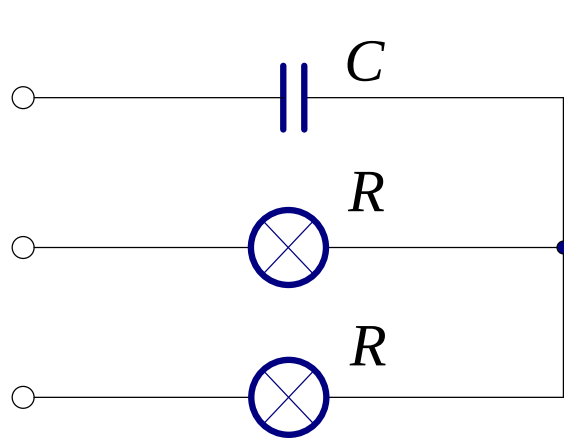
$$\Rightarrow \begin{cases} \dot{U}_{N_1N} = 27.3\angle 120^\circ \text{V} \\ \dot{U}_{N_2N} = 0 \end{cases}$$

$$\Rightarrow \dot{U}_{N_2N_1} = \dot{U}_{N_2N} - \dot{U}_{N_1N} = 27.3\angle -60^\circ \text{V}$$



例：图示不对称三相负载由构成相序测定器。当它的三个端点接至对称三相电源时，试说明如何根据两灯泡的明暗确定电源的相序。设 $R=(1/\omega C)$ 。

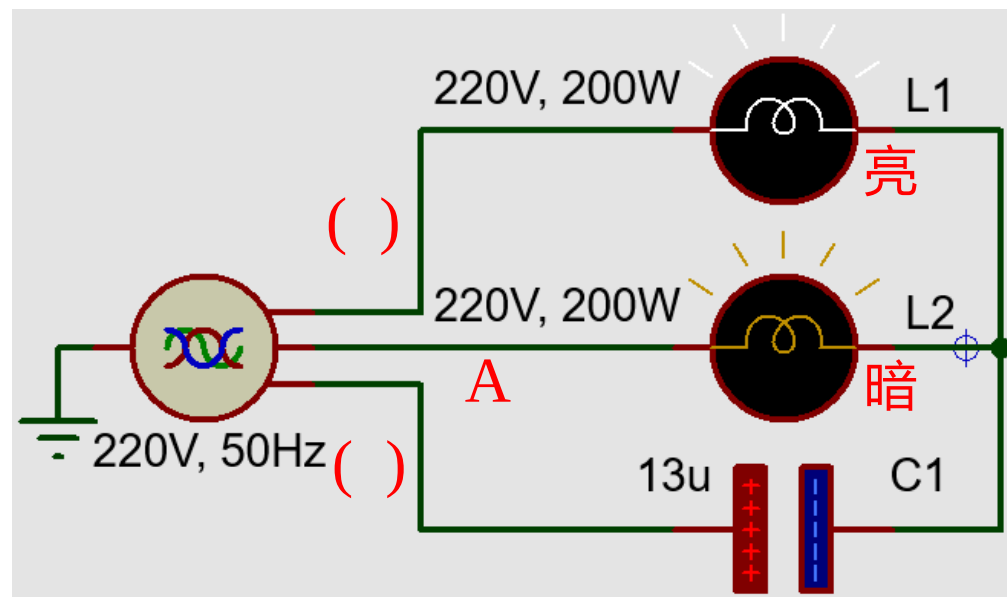
设 $\dot{U}_A = U \angle 0^\circ \text{ V}$



$$\dot{U}_{N'N} = \frac{j\omega C \dot{U}_A + \frac{\dot{U}_B}{R} + \frac{\dot{U}_C}{R}}{j\omega C + \frac{1}{R} + \frac{1}{R}} \approx (-0.2 + j0.6) \dot{U}_A$$

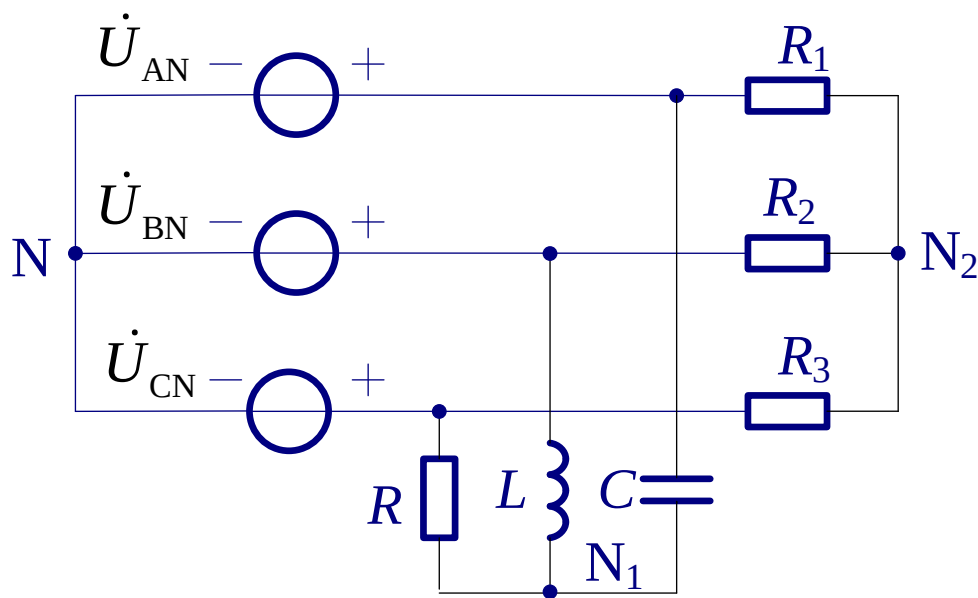
$$\dot{U}_{BN'} = \dot{U}_B - \dot{U}_{N'N} \approx 1.5 \angle -101.5^\circ \dot{U}_A$$

$$\dot{U}_{CN'} = \dot{U}_C - \dot{U}_{N'N} \approx 0.4 \angle 138^\circ \dot{U}_A$$



练习：下述正确的是

- A. N、 N_1 间的电压为零。
- B. N、 N_2 间的电压为零。
- C. 改变 R 、 L 、 C 的参数影响 R_1 支路的电流。
- D. 如果 $R_1 = R_2 = R_3$ ，则 N_1 、 N_2 间的电压为零。
- E. 两组三相负载可分开计算。



9.3 三相电路的功率

9.3.1 三相电路功率的基本概念

三相电路的瞬时功率 $p = p_A + p_B + p_C$

对称三相电路的瞬时功率

$$p_A = u_A i_A = U_{\text{pm}} \cos(\omega t) I_{\text{pm}} \cos(\omega t - \varphi) = U_p I_p [\cos \varphi + \cos(2\omega t - \varphi)]$$

$$p_B = U_p I_p [\cos \varphi + \cos(2\omega t - 240^\circ - \varphi)]$$

$$p_C = U_p I_p [\cos \varphi + \cos(2\omega t + 240^\circ - \varphi)]$$

$$\Rightarrow p = p_A + p_B + p_C = 3U_p I_p \cos \varphi = 3P_p = \text{constant}$$

如三相负载是电动机，则能获得稳定的转矩！

平均功率

$$P = P_A + P_B + P_C = U_{pA} I_{pA} \cos \varphi_A + U_{pB} I_{pB} \cos \varphi_B + U_{pC} I_{pC} \cos \varphi_C$$

无功功率

$$Q = Q_A + Q_B + Q_C = U_{pA} I_{pA} \sin \varphi_A + U_{pB} I_{pB} \sin \varphi_B + U_{pC} I_{pC} \sin \varphi_C$$

复功率

$$\tilde{S} = P + jQ$$

视在功率

$$S = \sqrt{P^2 + Q^2}$$

对称三相电路

$$\begin{cases} P = \sqrt{3} U_l I_l \cos \varphi \\ Q = \sqrt{3} U_l I_l \sin \varphi \end{cases} \quad \begin{cases} \tilde{S} = P + jQ \\ S = \sqrt{3} U_l I_l \\ S_A = S_B = S_C = \sqrt{3} U_l I_l / 3 \end{cases}$$

例：如图，已知 $f=50\text{Hz}$ ，负载阻抗 $Z=(18+j24)\Omega$ 。为使功率因数提高到 0.9，在负载端口处接入星形连接的电容元件组，试求接入的每相电容 C 的值。

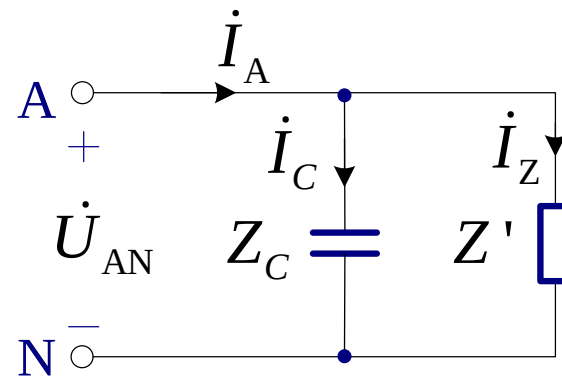
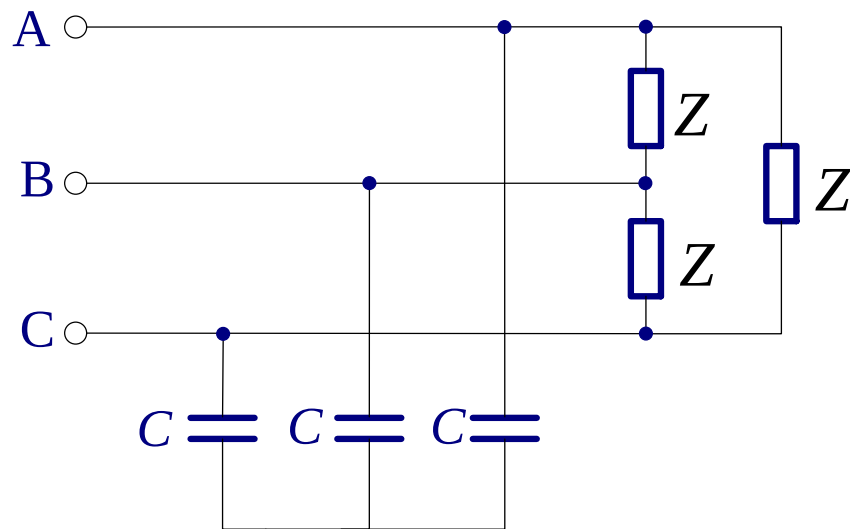
$$\cos \varphi = 0.6 \Rightarrow \varphi = 53.13^\circ$$

$$\cos \varphi' = 0.9 \Rightarrow \varphi' = 25.84^\circ$$

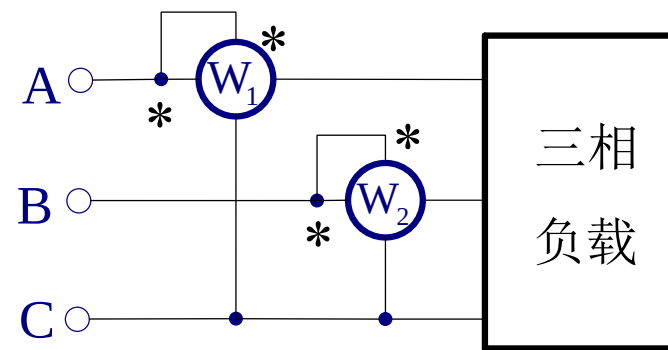
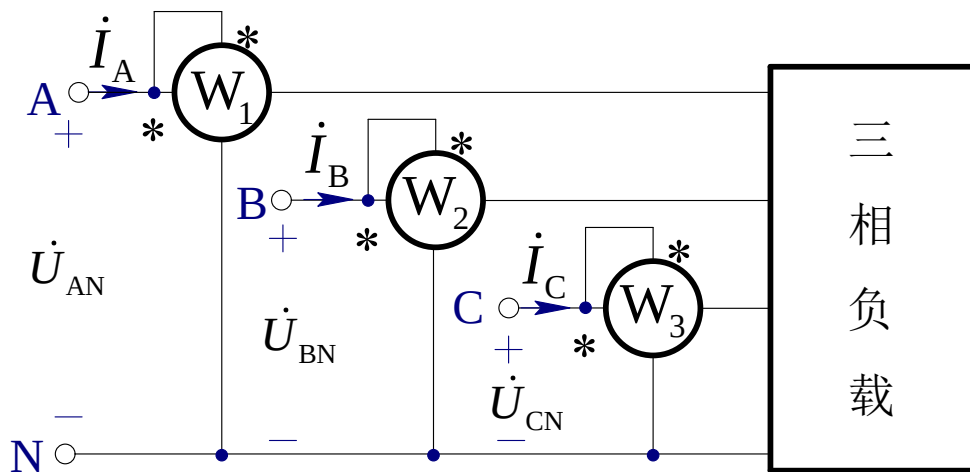
$$P_A = U_{AN} I_Z \cos \varphi = \frac{U_{AN}^2}{|Z'|} \cos \varphi$$

$$C = \frac{P_A}{\omega U_{AN}^2} (\tan \varphi - \tan \varphi')$$

$$= \frac{\cos \varphi}{\omega |Z'|} (\tan \varphi - \tan \varphi') = 1.62 \times 10^{-4} \text{ F}$$



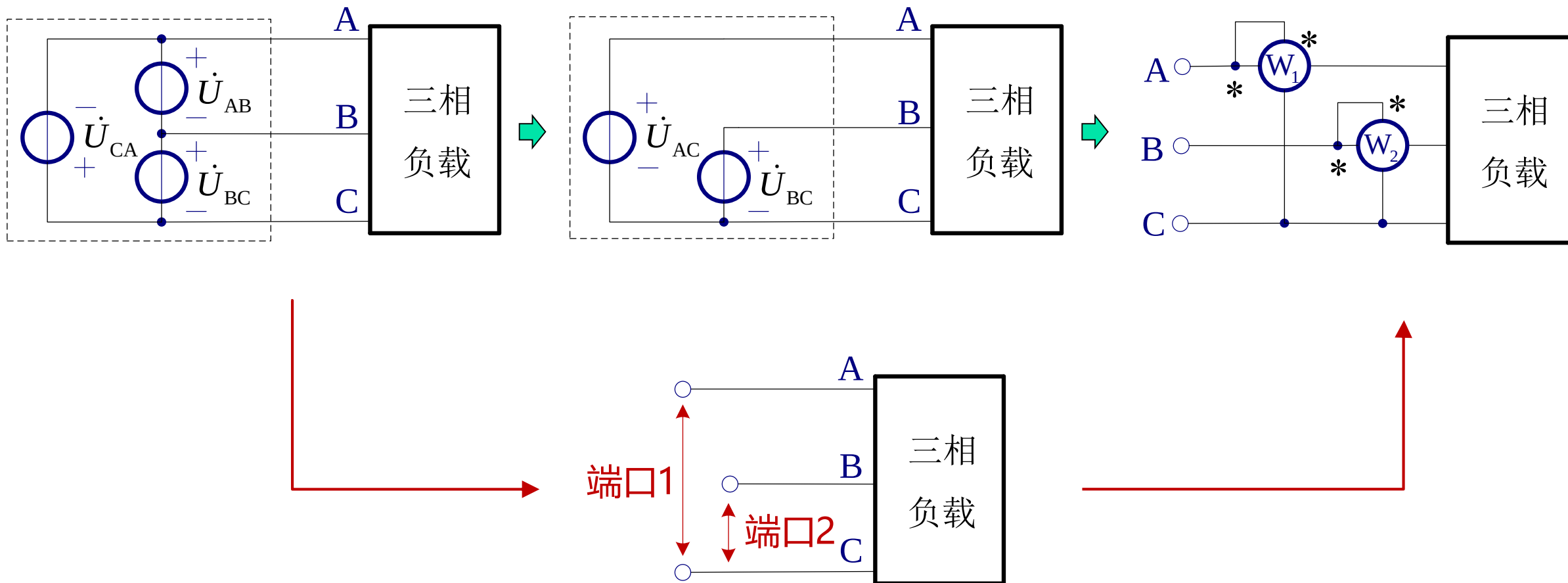
9.3.2 三相电路功率的测量



两功率表法

$$\begin{aligned}
 P_1 + P_2 &= \operatorname{Re}[\dot{U}_{AC} \dot{I}_A^*] + \operatorname{Re}[\dot{U}_{BC} \dot{I}_B^*] = \operatorname{Re}[\dot{U}_{AC} \dot{I}_A^* + \dot{U}_{BC} \dot{I}_B^*] \\
 &= \operatorname{Re}[(\dot{U}_{AN} - \dot{U}_{CN}) \dot{I}_A^* + (\dot{U}_{BN} - \dot{U}_{CN}) \dot{I}_B^*] \\
 &= \operatorname{Re}[\dot{U}_{AN} \dot{I}_A^* + \dot{U}_{BN} \dot{I}_B^* - \dot{U}_{CN} (\dot{I}_A^* + \dot{I}_B^*)] \\
 &= \operatorname{Re}[\dot{U}_{AN} \dot{I}_A^* + \dot{U}_{BN} \dot{I}_B^* + \dot{U}_{CN} \dot{I}_C^*] \\
 &= U_{AN} I_A \cos \varphi_A + U_{BN} I_B \cos \varphi_B + U_{CN} I_C \cos \varphi_C = P_A + P_B + P_C = P
 \end{aligned}$$

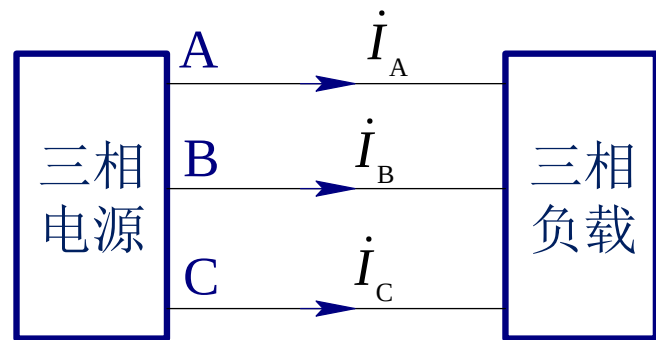
两功率表法的直观理解



证明：对如图三相三线制电路中的任一节点J，则三相负载吸收的功率为

$$P = \operatorname{Re}[\dot{U}_{AJ} \dot{I}_A^*] + \operatorname{Re}[\dot{U}_{BJ} \dot{I}_B^*] + \operatorname{Re}[\dot{U}_{CJ} \dot{I}_C^*]$$

证：假设三相电源的中点为N，则有



$$P = \operatorname{Re}[\dot{U}_{AN} \dot{I}_A^*] + \operatorname{Re}[\dot{U}_{BN} \dot{I}_B^*] + \operatorname{Re}[\dot{U}_{CN} \dot{I}_C^*]$$

$$= \operatorname{Re}[\dot{U}_{AN} \dot{I}_A^*] + \operatorname{Re}[\dot{U}_{BN} \dot{I}_B^*] + \operatorname{Re}[\dot{U}_{CN} \dot{I}_C^*] - \operatorname{Re}[\dot{U}_{JN} (\dot{I}_A^* + \dot{I}_B^* + \dot{I}_C^*)]$$

$$\dot{I}_A^* + \dot{I}_B^* + \dot{I}_C^* = 0$$

$$= \operatorname{Re}[(\dot{U}_{AN} - \dot{U}_{JN}) \dot{I}_A^*] + \operatorname{Re}[(\dot{U}_{BN} - \dot{U}_{JN}) \dot{I}_B^*] + \operatorname{Re}[(\dot{U}_{CN} - \dot{U}_{JN}) \dot{I}_C^*]$$

$$= \operatorname{Re}[\dot{U}_{AJ} \dot{I}_A^*] + \operatorname{Re}[\dot{U}_{BJ} \dot{I}_B^*] + \operatorname{Re}[\dot{U}_{CJ} \dot{I}_C^*]$$

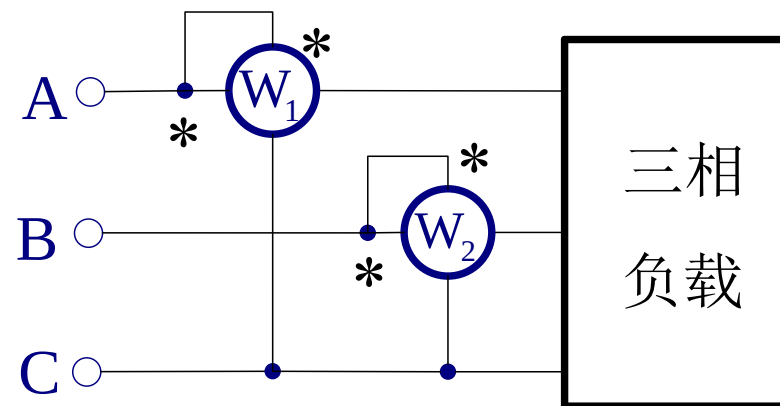
讨论：

1. 选节点J为A、B或C，得到两功率表法。
2. 上述结论对三相四线制电路是否成立？

练习：下述正确的是

- A. 两功率表法应用时与电路的相序无关。
- B. 两功率表法应用时与电路的连接形式是否是三角形联结还是星形联结有关。
- C. 两功率表法应用时必须将两个功率表同时接入电路。
- D. 两功率表法应用时与电路的是否对称有关。
- E. 两功率表法在某些三相三线制电路中失效。

例：在对称负载的情况下，三相电源正序，已知两功率表的读数 P_1 、 P_2 ，试求负载的无功功率和阻抗角。



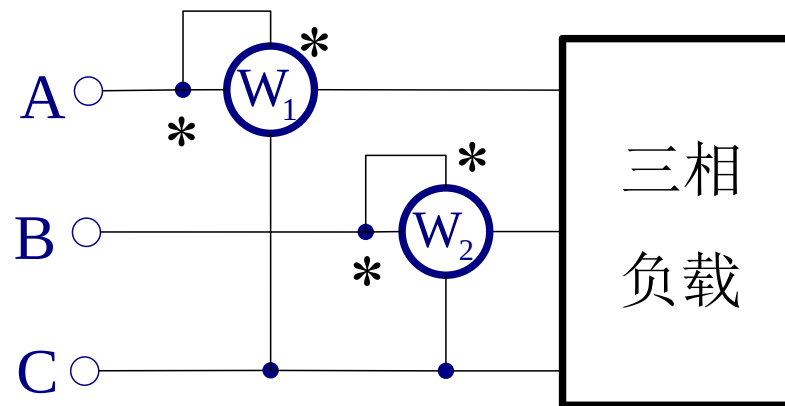
问题分析：

1. 列出关于无功功率和阻抗角的两个方程；
2. 如何列写：将 P_1 、 P_2 用无功功率和阻抗角表达出来！

例：在对称负载的情况下，三相电源正序，已知两功率表的读数 P_1 、 P_2 ，试求负载的无功功率和阻抗角。

为方便分析，设负载为星形联接，A相电压初相为0，则线电压为

$$\dot{U}_{AB} = U_l \angle 30^\circ \quad \dot{U}_{BC} = U_l \angle -90^\circ \quad \dot{U}_{CA} = U_l \angle 150^\circ$$



线电流可表示为 $\dot{I}_A = I_l \angle -\varphi_Z$ $\dot{I}_B = I_l \angle (-\varphi_Z - 120^\circ)$

$$\begin{aligned} P_1 &= \text{Re}[\dot{U}_{AC} \dot{I}_A^*] = \text{Re}[-\dot{U}_{CA} \dot{I}_A^*] = \text{Re}[-U_l \angle 150^\circ \times I_l \angle \varphi_Z] \\ &= \text{Re}[U_l \angle -30^\circ \times I_l \angle \varphi_Z] = \text{Re}[U_l I_l \angle (\varphi_Z - 30^\circ)] = U_l I_l \cos(\varphi_Z - 30^\circ) \end{aligned}$$

$$P_2 = \text{Re}[\dot{U}_{BC} \dot{I}_B^*] = \text{Re}[U_l \angle -90^\circ \times I_l \angle (\varphi_Z + 120^\circ)] = \text{Re}[U_l I_l \angle (\varphi_Z + 30^\circ)] = U_l I_l \cos(\varphi_Z + 30^\circ)$$

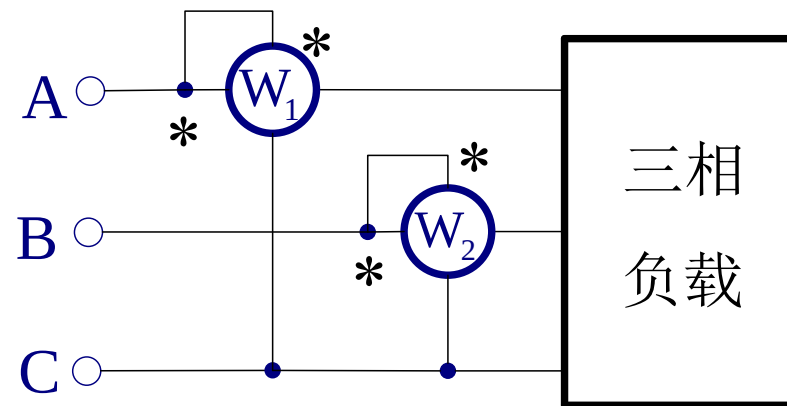
例：在对称负载的情况下，三相电源正序，已知两功率表的读数 P_1 、 P_2 ，试求负载的无功功率和阻抗角。

$$\begin{cases} P_1 = U_l I_l \cos(\varphi_Z - 30^\circ) \\ P_2 = U_l I_l \cos(\varphi_Z + 30^\circ) \end{cases}$$

$$\Rightarrow \begin{cases} P_1 = \frac{\sqrt{3}}{2} U_l I_l \cos \varphi_Z + \frac{1}{2} U_l I_l \sin \varphi_Z \\ P_2 = \frac{\sqrt{3}}{2} U_l I_l \cos \varphi_Z - \frac{1}{2} U_l I_l \sin \varphi_Z \end{cases}$$

$$P_1 - P_2 = U_l I_l \sin \varphi_Z$$

$$Q = \sqrt{3}(P_1 - P_2) \quad \varphi_Z = \arctan \frac{Q}{P} = \arctan \sqrt{3} \frac{P_1 - P_2}{P_1 + P_2}$$



思考：三相电源逆序的情况？

例：如图，已知线电压 U 、电阻 R ，试求功率表的读数。

设 $\dot{U}_{AB} = U \angle 0^\circ \text{ V}$

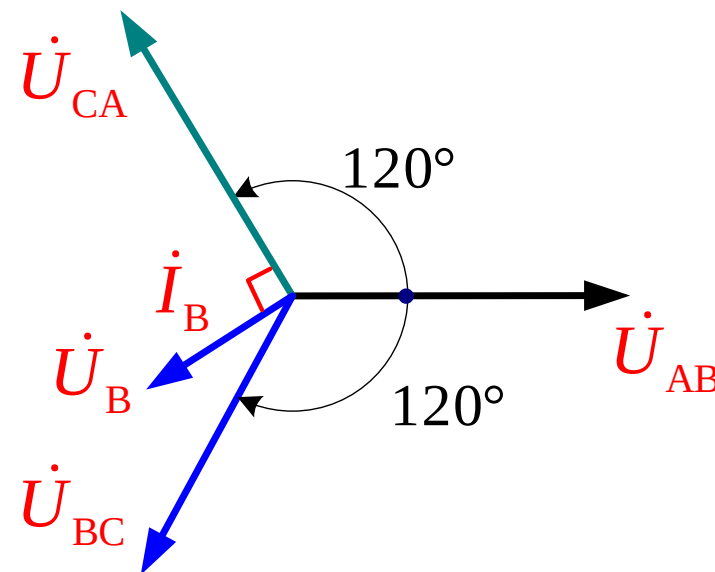
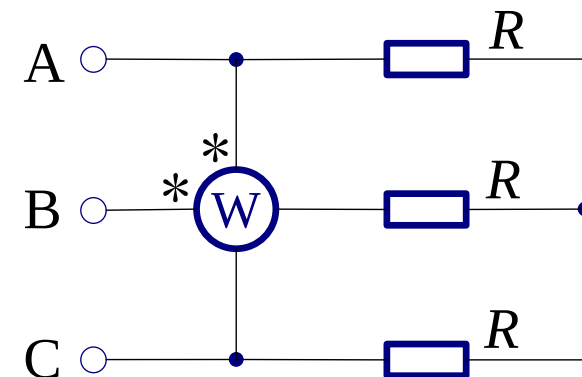
$$\Rightarrow \begin{cases} \dot{U}_{AC} = -\dot{U}_{CA} = U \angle (120^\circ - 180^\circ) \text{ V} \\ \dot{I}_B = \dot{U}_B / R = (\dot{U}_{BC} / \sqrt{3}) \angle -30^\circ / R = (U / \sqrt{3}) \angle (-120^\circ - 30^\circ) / R \end{cases}$$

$$P = \text{Re}[\dot{U}_{AC} \dot{I}_B^*]$$

$$= \text{Re}[U \angle (120^\circ - 180^\circ) \times \left[\frac{(U / \sqrt{3}) \angle (-120^\circ - 30^\circ)}{R} \right]^*]$$

$$= \text{Re}\left[\frac{U^2}{\sqrt{3}R} \angle (-60^\circ + 150^\circ) \right] = 0$$

正交的电、电压、电流不产生平均功率！



例：求两功率表的读数。

设

$$\begin{cases} \dot{U}_{AB} = 45\angle 0^\circ \text{ V} \\ \dot{U}_{BC} = 45\angle -120^\circ \text{ V} \\ \dot{U}_{CA} = 45\angle 120^\circ \text{ V} \end{cases}$$

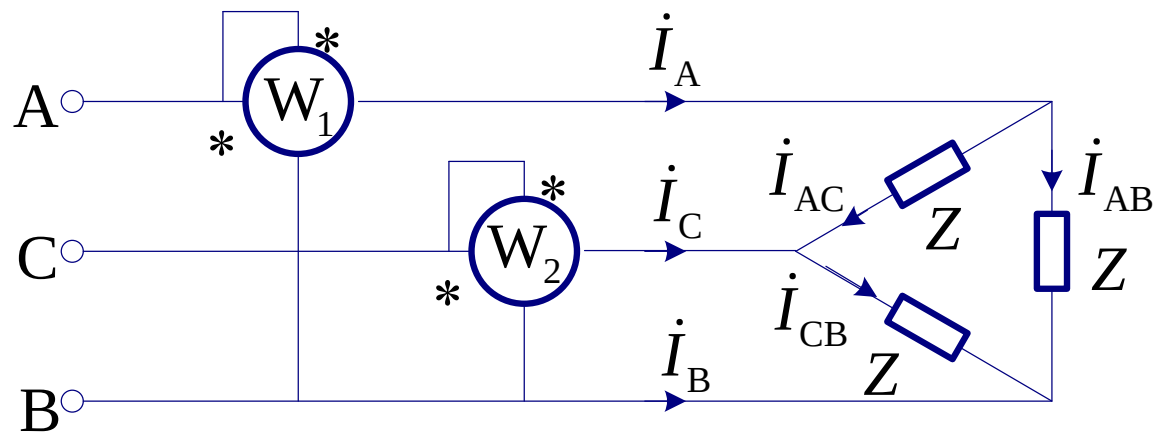
$$\dot{I}_A = \dot{U}_{AB}/Z + \dot{U}_{AC}/Z$$

$$= (45\angle 0^\circ - 45\angle 120^\circ)/(9 - j12) = (4.7785 + j2.0412)\text{A}$$

$$\dot{U}_{AB}\dot{I}_A^* = 45\angle 0^\circ \times (4.7785 - j2.0412) = (215.03 - j91.85)\text{VA}$$

$$W_1 = 215\text{W}$$

$$W_2 = 28\text{W}$$



$$Z = (9 - j12)\Omega, U_l = 45\text{V}, \text{正相序}$$

例：如图，线电压为380V，接有两组对称三相负载，已知星形负载的每组负载阻抗为 $Z=(30+j40)\Omega$ ，电动机负载的功率为 $P_M=1700W$ ，功率因数为0.8（感性）。试求三个功率表的示数。

分析：由于负载对称，可用一相分析法。

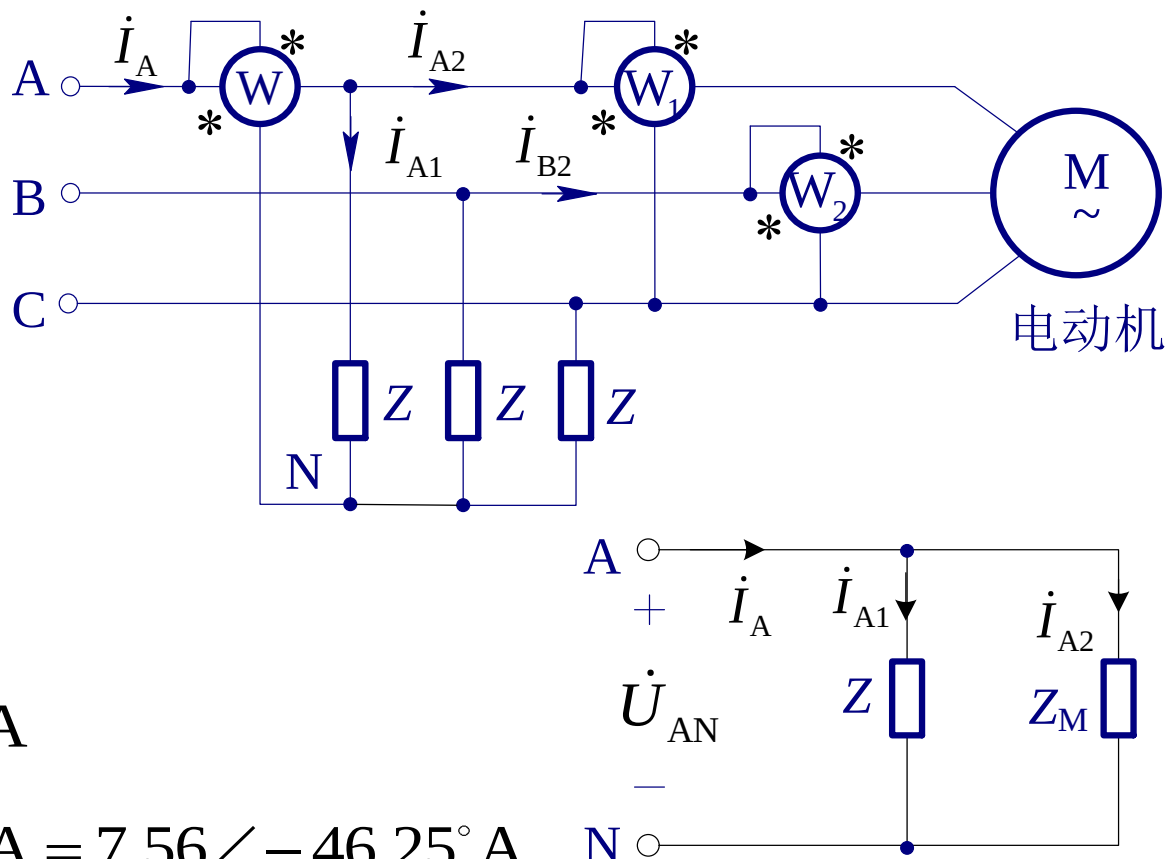
$$\dot{U}_{AN} = 220\angle 0^\circ \text{ V}$$

$$\dot{I}_{A1} = \frac{\dot{U}_{AN}}{Z} = \frac{220\angle 0^\circ}{30 + j40} \text{ A} = 4.4\angle -53.1^\circ \text{ A}$$

$$I_{A2} = \frac{P_M}{\sqrt{3}U_l \cos \varphi} = \frac{1700}{\sqrt{3} \times 380 \times 0.8} \text{ A} = 3.23 \text{ A}$$

$$\cos \varphi = 0.8 \Rightarrow \varphi = 36.9^\circ \quad \dot{I}_{A2} = 3.23\angle -36.9^\circ \text{ A}$$

$$\dot{I}_A = \dot{I}_{A1} + \dot{I}_{A2} = (4.4\angle -53.1^\circ + 3.23\angle -36.9^\circ) \text{ A} = 7.56\angle -46.25^\circ \text{ A}$$

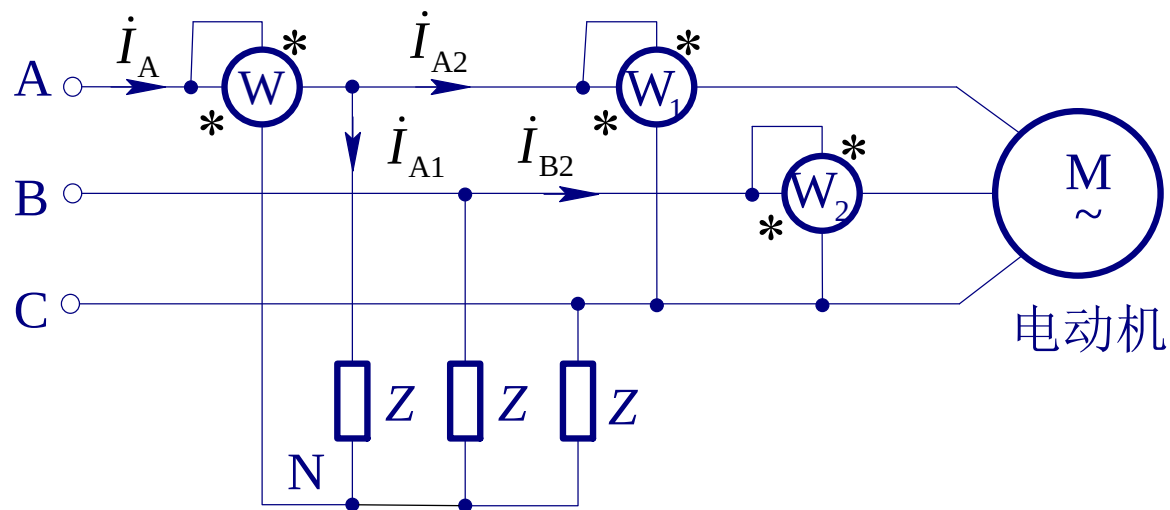


$$\dot{U}_{AN} = 220\angle 0^\circ \text{ V}$$

$$\dot{I}_{A1} = 4.4\angle -53.1^\circ \text{ A}$$

$$\dot{I}_{A2} = 3.23\angle -36.9^\circ \text{ A}$$

$$\dot{I}_A = \dot{I}_{A1} + \dot{I}_{A2} = 7.56\angle -46.25^\circ \text{ A}$$



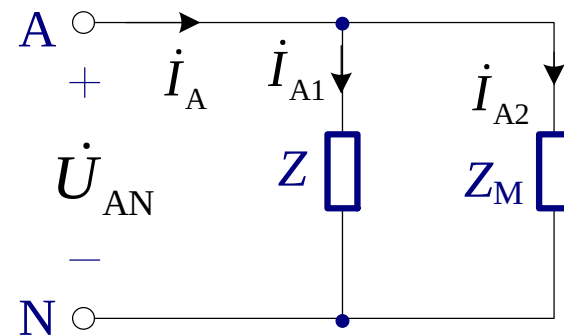
$$P = \text{Re}[\dot{U}_{AN} \dot{I}_A^*]$$

$$= \text{Re}[220\angle 0^\circ \times 7.56\angle 46.25^\circ] \text{ W} = 1150 \text{ W}$$

$$P_1 = U_{AC} I_{A2} \cos[-30^\circ - (-36.9^\circ)]$$

$$= 380 \times 3.23 \cos(36.9^\circ - 30^\circ) \text{ W} = 1218 \text{ W}$$

$$P_2 = U_{BC} I_{B2} \cos[(-120^\circ + 30^\circ) - (-36.9^\circ - 120^\circ)] = 380 \times 3.23 \cos(66.9^\circ) \text{ W} = 482 \text{ W}$$

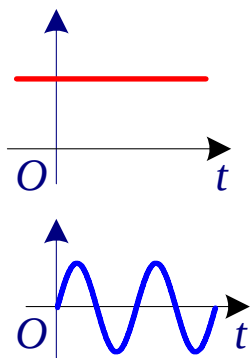


练习：请选择正确的表述。

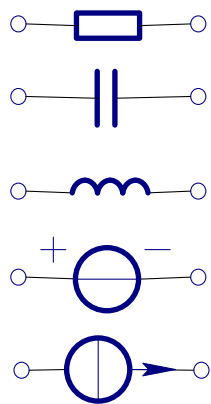
- A. 不对称三相电路也可采用一相分析法求解。
- B. 三相电动机的铭牌上给出的额定功率就是电源输送给电动机的有功功率。
- C. 三相电路中，三相负载的功率因数角就是其各相的阻抗角。
- D. 给三相感性负载并联电容并不能提高感性负载的功率因数。
- E. 只有在三相三线制电路中，才能用两功率表法测量三相电路的有功功率。

课程总结

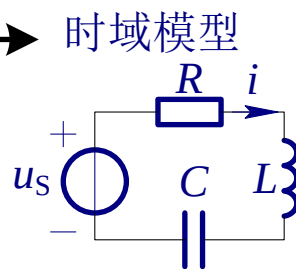
① 基本波形



② 电路元件



③ 电路模型



④ 两类约束

拓扑约束 元件约束

时域形式

$$\begin{aligned} \sum u &= 0 & u &= R i \\ \sum i &= 0 & u &= L \frac{di}{dt} \\ & & i &= C \frac{du}{dt} \end{aligned}$$

⑤ 分析方法

网孔分析法
节点分析法
回路分析法
割集分析法
等效变换
电路定理
三要素法

激励

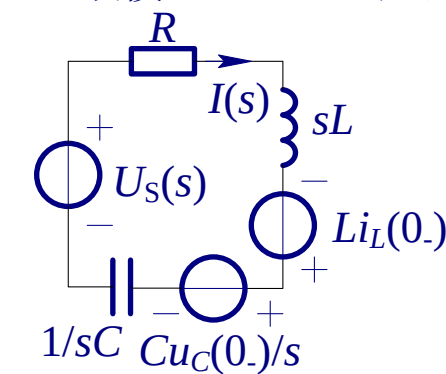
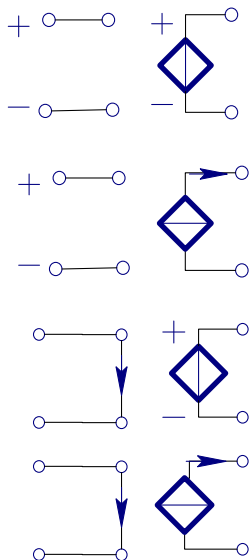
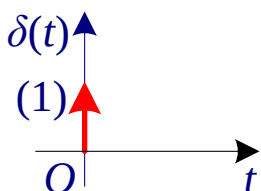
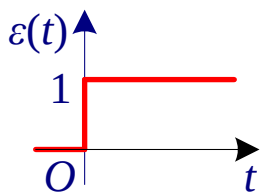
电路

s域模型

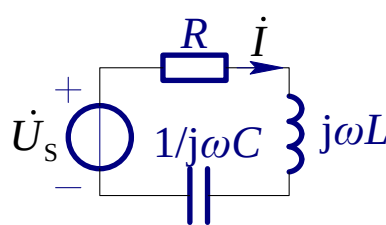
电路方程

全响应

稳态响应



相量模型



s域形式

$$\begin{aligned} \sum U(s) &= 0 & U(s) &= Z(s)I(s) \\ \sum I(s) &= 0 \end{aligned}$$

相量形式

$$\begin{aligned} \sum \dot{U} &= 0 & \dot{U} &= Z\dot{I} \\ \sum \dot{I} &= 0 \end{aligned}$$

拉氏变换法
相量分析法